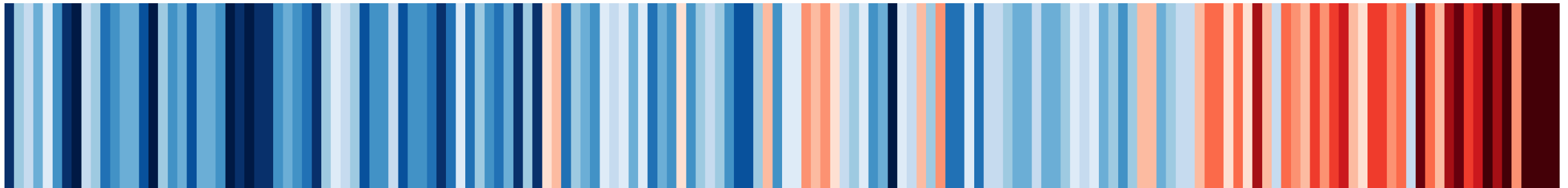


Psychology of Climate Change Communication

Week 3: A question of trust

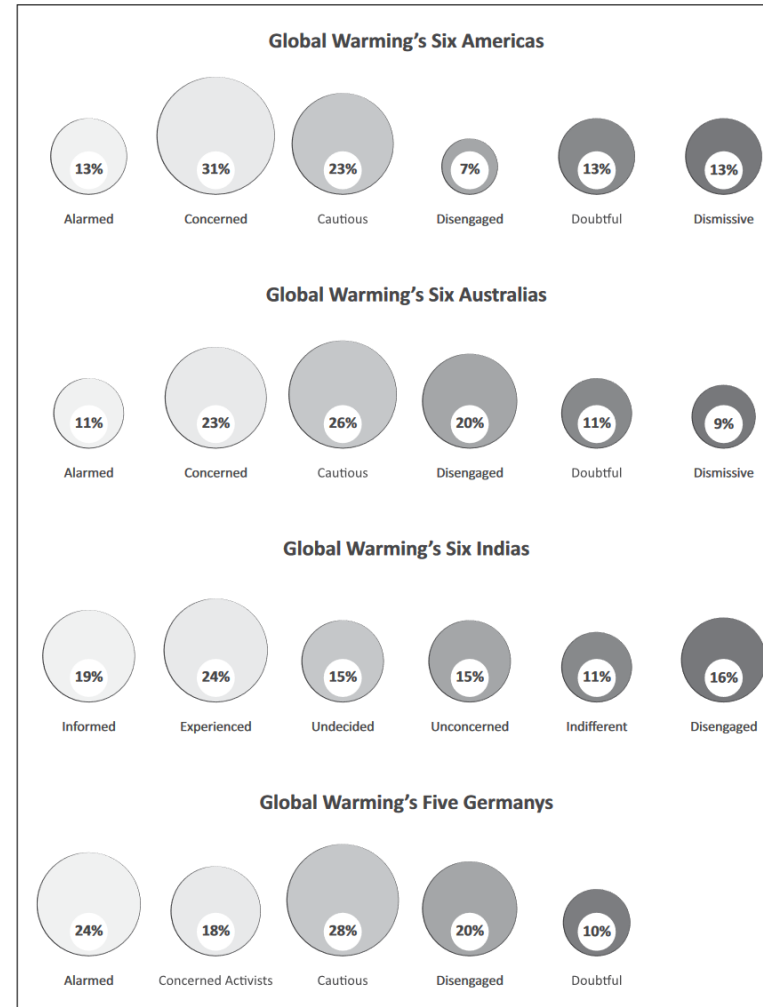
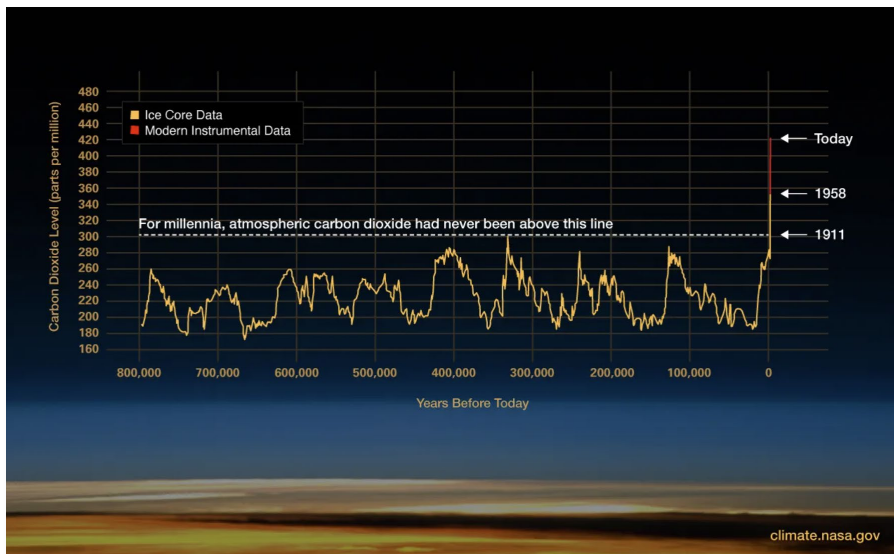
Dr. Viktoria Cologna and Dr. Jan Pfänder



Let's recap

- Public opinion data help communicators understand the size of different audiences
- People underestimate the scientific consensus on climate change and other people's support for climate action (pluralistic ignorance)
 - Communicating about the scientific consensus can increase belief and worry
- The importance of knowledge for climate action depends on the type of knowledge
 - Communicate facts about the causes and consequences
- People systematically underestimate the impact of high-impact behaviors
 - Communication on behaviors' impact can increase engagement in these behaviors, but potentially backfire on collective action

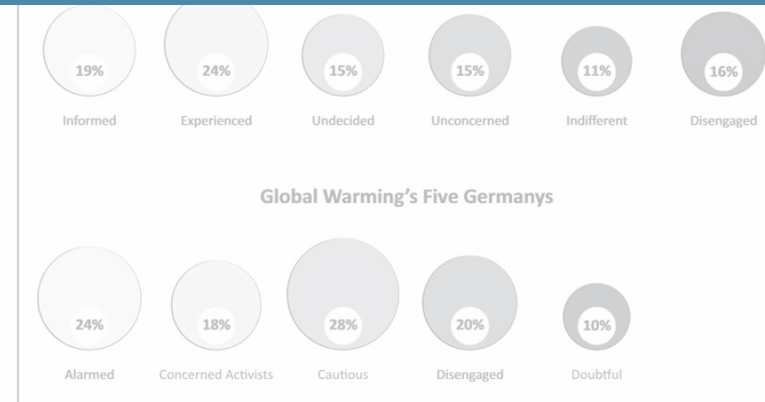
A question of trust



A question of trust



Communication strategies are likely to fail
if they don't come from a *trusted messenger*.



The Messenger effect

People judge the validity or relevance of information based on the person delivering the message.

- The weight we give to information depends greatly on the automatic reactions we have to the perceived authority of the source of that information – the ‘messenger’ (Dolan et al., 2012).
- Instead of objectively analyzing the message’s content, our opinions of the person delivering the information affect our interpretation.
- Information from trusted sources is more likely to be attended to, considered, accepted, and acted upon (Maibach et al., 2023).

→ A message is only as effective as the person delivering it.

A question of trust

- What makes someone trustworthy?
- Who are trusted messengers?

Definition of trust

Trust is “a psychological state comprising the intention to accept vulnerability based upon positive expectations of the intentions or behaviour of another”. (Rousseau et al., 1998, p. 395).

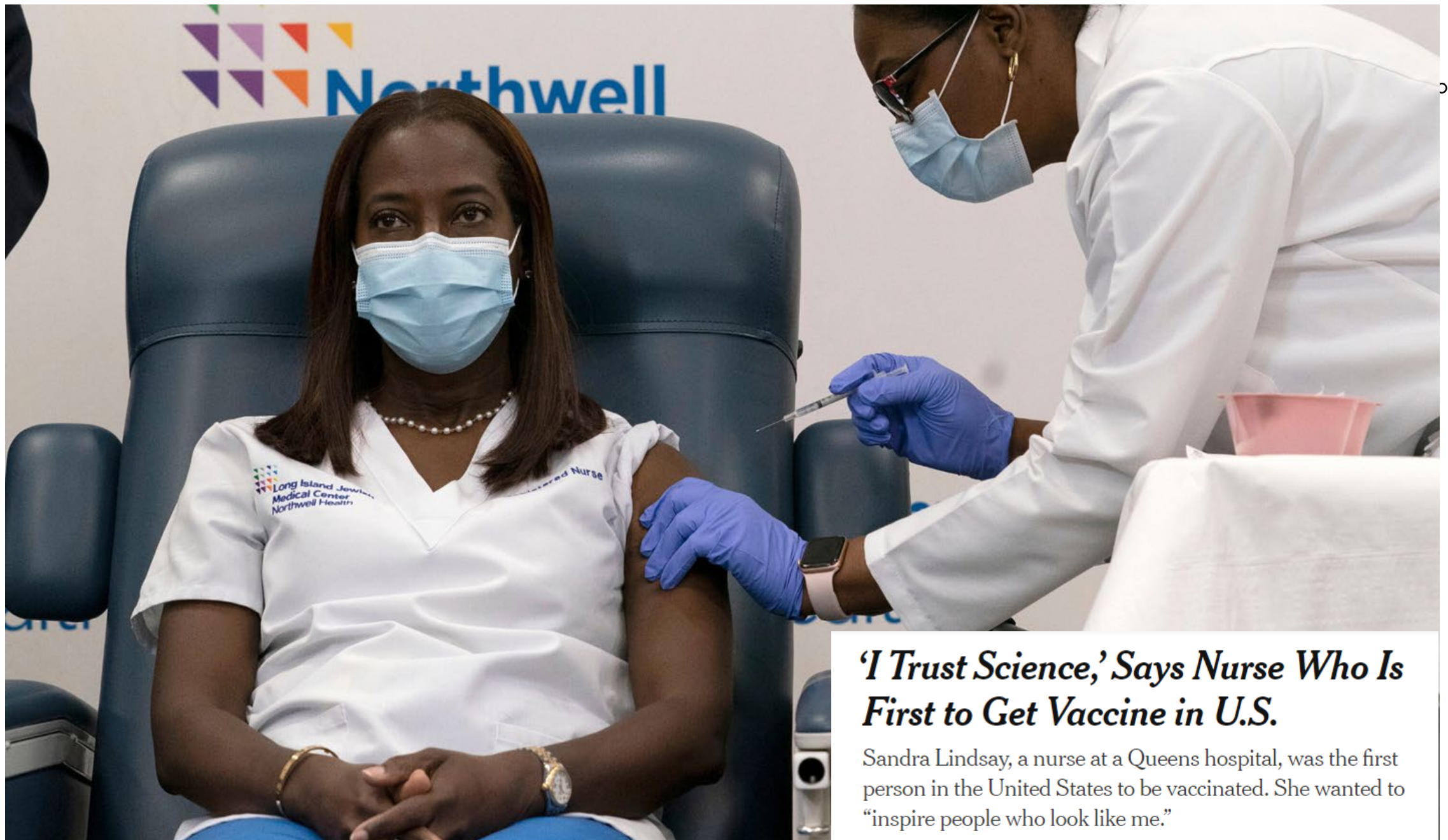
What is trust?



<https://www.flickr.com/photos/194798290@N03/53345322284/>

What is trust?

- Trust is essential in navigating the complexity of modern societies (Luhmann, 1989)
- Trust in others reduces the risk of having to rely on the unknown and enables us to benefit from the expertise of others (Goodwin & Dahlstrom, 2014)
- Trust makes us vulnerable to the risk that others will not act as expected (Lucas et al., 2015)
- Trust helps individuals make decisions, especially when information is low and uncertainties high (Earle, 2010; Siegrist, 2021)



'I Trust Science,' Says Nurse Who Is First to Get Vaccine in U.S.

Sandra Lindsay, a nurse at a Queens hospital, was the first person in the United States to be vaccinated. She wanted to "inspire people who look like me."



“Trust in scientists
is the critical determinant of
societies’ resilience in their
fight against the pandemic.”

Algan et al., 2021.

Trust depends on...

- The person who trusts.
- The person or institution who is to be trusted.
- The context: the matter in hand and the circumstances around it.

The mix of these three factors determines whether or not people are able to trust in any given moment.

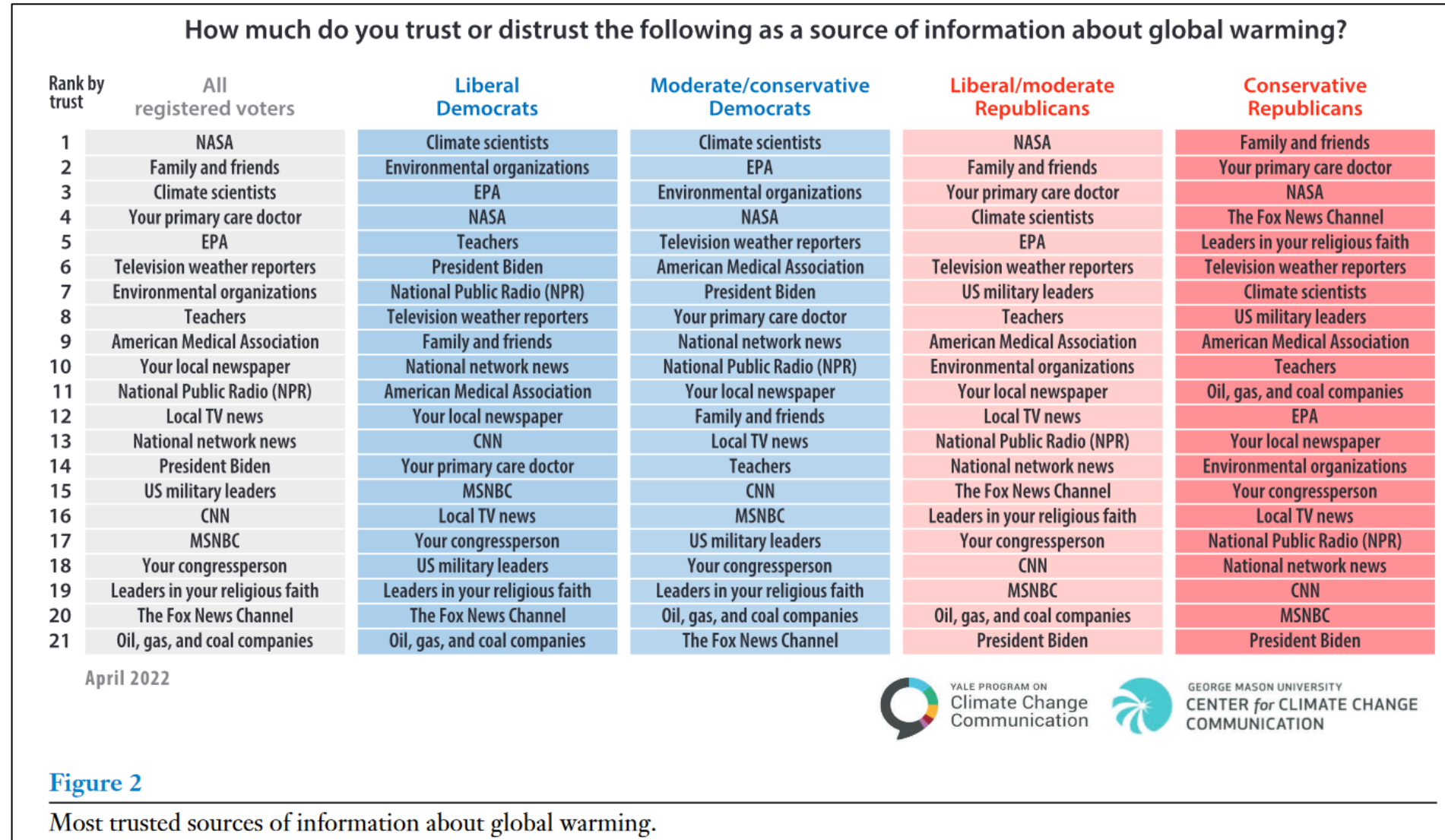
Climate Outreach, 2024

Trust and climate change

- Some people are trusted more when it comes to climate change.
- Certain public figures and groups embody the characteristics people find trustworthy.
- They have a vital role to play in communicating about climate change.

What is important to know:

- Which celebrities, politicians, professions or other interest groups are more trusted by certain audiences?
- Who are trusted messengers?
- What makes us trust these people?



Mentimeter

Why makes a climate change communicator trustworthy?

What explains perceived trustworthiness?

Trustworthiness perceptions are influenced by (Mayer et al., 1995):

- Ability: Seeing someone as having knowledge, skills, and competency
- Integrity: Seeing someone as honest and sincere.
- Benevolence: Seeing someone as possessing goodwill and having our best interests in mind

None of these factors are inherently objective.

- Who we see as having expertise and integrity, and who we assume to have our best interests at heart is highly subjective.

What explains perceived trustworthiness?

- Perceptions of a messenger may also depend on perceptions of competence and warmth.

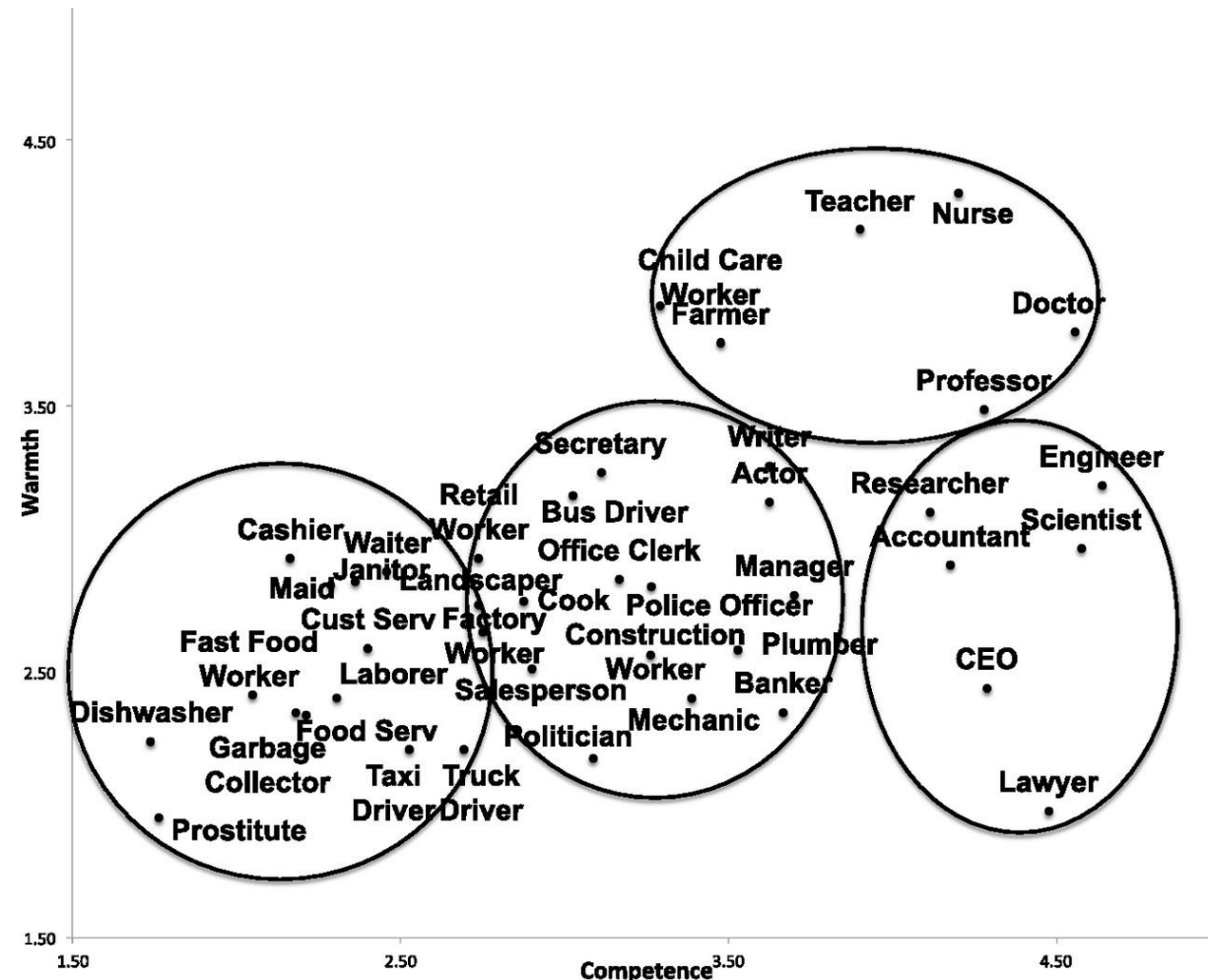
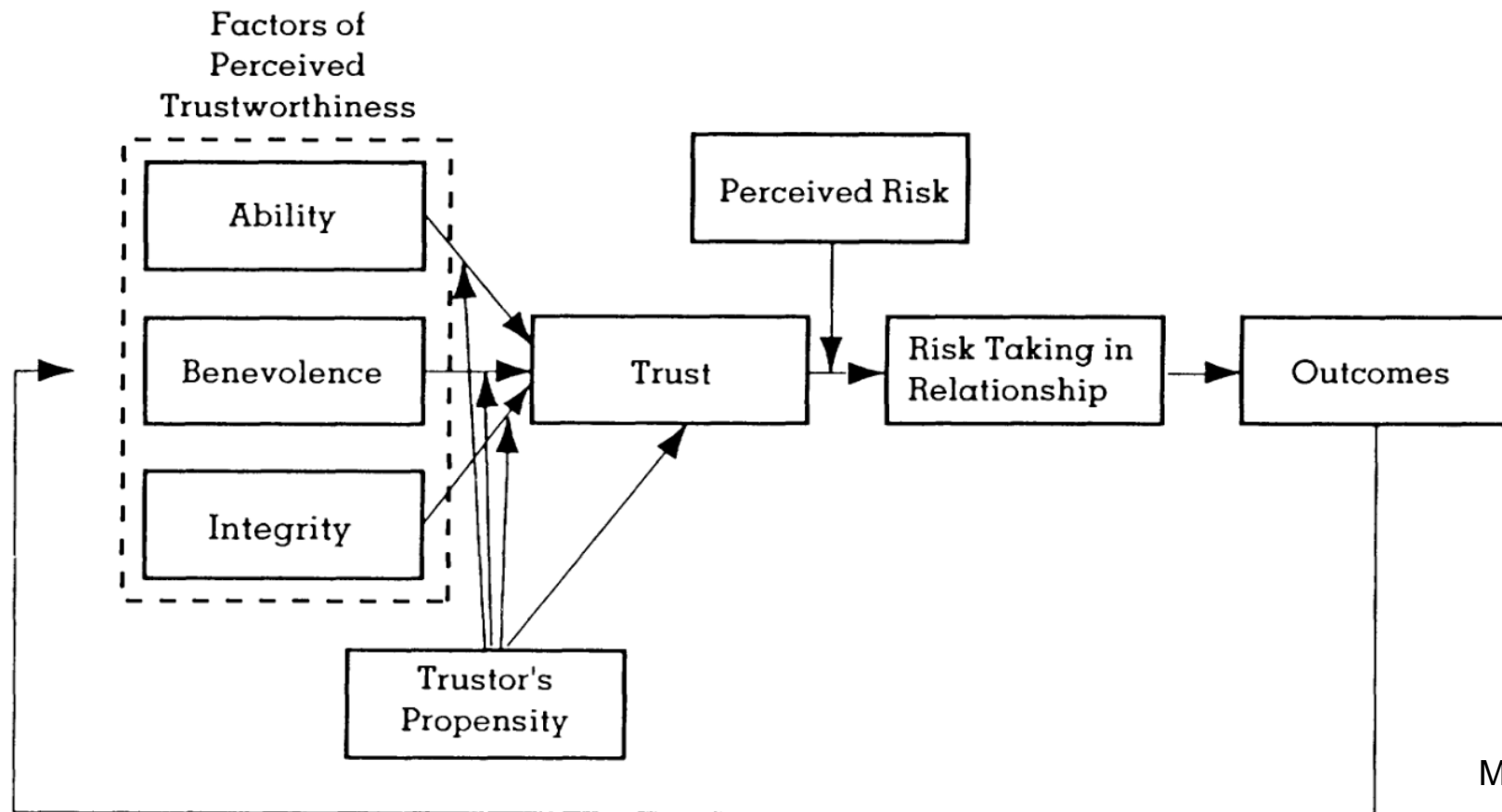


FIGURE 1
Proposed Model of Trust



Mayer et al., 1995

What explains perceived trustworthiness?

- Value similarity: the degree to which a person believes their personal values and goals match those of other individuals, social groups, or governing institutions.
 - People trust similar others, people they perceive as being in their own group, and have similar values and identities.
- Attractive people are viewed as more trustworthy and are more likely to benefit from the trust of others.

Trusted messengers: Weathercasters

Broadcast meteorologists are uniquely positioned to educate the public about the personal relevance of climate change:

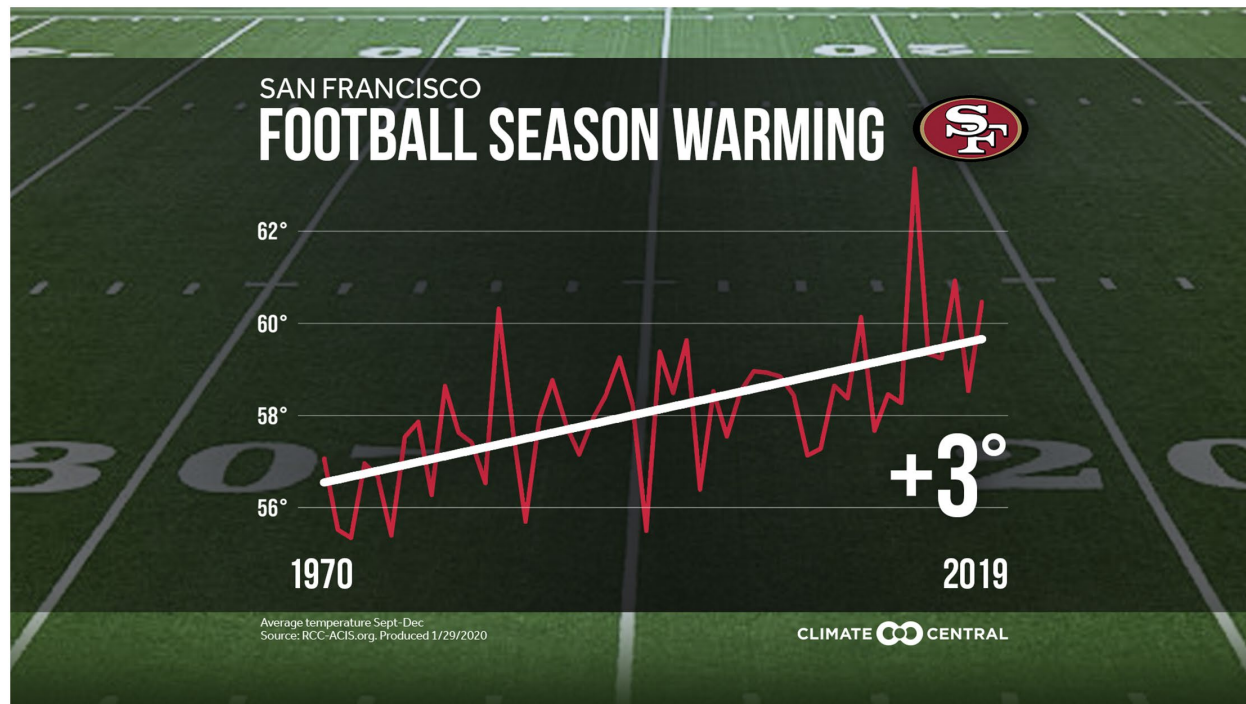
- high public trust as a source of climate science information
- local TV is the #1 source of weather information in America
- most weathercasters have relevant scientific training and excellent communication skills

Surveys show that most TV meteorologists would like to report on climate change, but lack of time, broadcast-quality graphics, and access to appropriate experts.

The Climate Matters Program

A one-year field experiment conducted in 2010 with the TV station WLTX (South Carolina) tested the premise that weathercasters are well positioned to educate audiences about the local relevance of global climate change.

The Climate Matters Program



The Climate Matters Program

<https://www.climatecentral.org/climate-matters>

The Climate Matters Program

- Viewers of Climate Matters were more likely to hold a range of science-based beliefs about climate change (Zhao et al., 2014)
- Significant positive association between the amount of Climate Matters reporting and some key indicators of science-based understanding (including that climate change is occurring, is primarily human caused, and causes harm) (Meyers et al., 2020)
- Climate Matters produces free weekly climate reporting materials in English and Spanish, localized for 245+ U.S. cities and media markets.

Trusted messengers: Climate scientists

How much do you trust or distrust the following as a source of information about global warming?

Rank by trust	All registered voters	Liberal Democrats	Moderate/conservative Democrats	Liberal/moderate Republicans	Conservative Republicans
1	NASA	Climate scientists	Climate scientists	NASA	Family and friends
2	Family and friends	Environmental organizations	EPA	Family and friends	Your primary care doctor
3	Climate scientists	EPA	Environmental organizations	Your primary care doctor	NASA
4	Your primary care doctor	NASA	NASA	Climate scientists	The Fox News Channel
5	EPA	Teachers	Television weather reporters	EPA	Leaders in your religious faith
6	Television weather reporters	President Biden	American Medical Association	Television weather reporters	Television weather reporters
7	Environmental organizations	National Public Radio (NPR)	President Biden	US military leaders	Climate scientists
8	Teachers	Television weather reporters	Your primary care doctor	Teachers	US military leaders
9	American Medical Association	Family and friends	National network news	American Medical Association	American Medical Association
10	Your local newspaper	National network news	National Public Radio (NPR)	Environmental organizations	Teachers
11	National Public Radio (NPR)	American Medical Association	Your local newspaper	Your local newspaper	Oil, gas, and coal companies
12	Local TV news	Your local newspaper	Family and friends	Local TV news	EPA
13	National network news	CNN	Local TV news	National Public Radio (NPR)	Your local newspaper
14	President Biden	Your primary care doctor	Teachers	National network news	Environmental organizations
15	US military leaders	MSNBC	CNN	The Fox News Channel	Your congressperson
16	CNN	Local TV news	MSNBC	Leaders in your religious faith	Local TV news
17	MSNBC	Your congressperson	US military leaders	Your congressperson	National Public Radio (NPR)
18	Your congressperson	US military leaders	Your congressperson	CNN	National network news
19	Leaders in your religious faith	Leaders in your religious faith	Leaders in your religious faith	MSNBC	CNN
20	The Fox News Channel	The Fox News Channel	Oil, gas, and coal companies	Oil, gas, and coal companies	MSNBC
21	Oil, gas, and coal companies	Oil, gas, and coal companies	The Fox News Channel	President Biden	President Biden

April 2022



Figure 2

Most trusted sources of information about global warming.

Trust in climate scientists

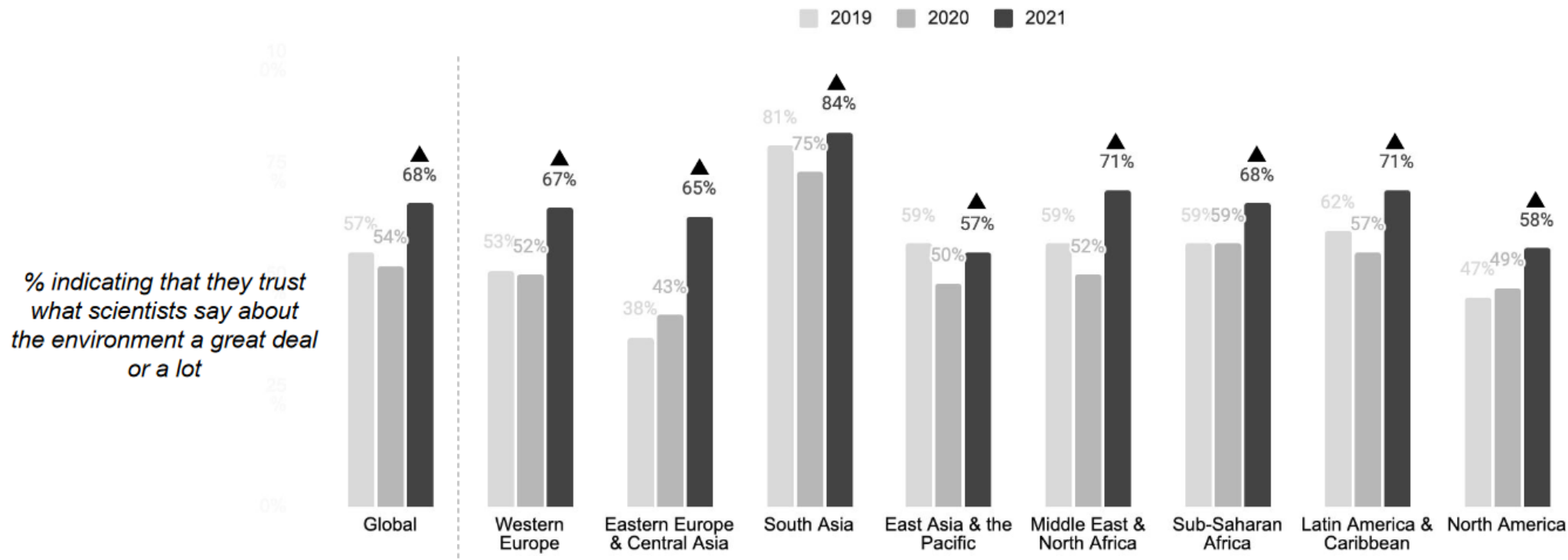
- People with higher trust in scientists are more likely to accept the scientific consensus on global warming, have more accurate beliefs about climate change, and support climate policies (Ejaz, Vu, and Fletcher 2025; Bogert et al. 2024; Cologna and Siegrist 2020; Hornsey et al. 2016).
- A study across 55 countries found that trust in climate scientists was the strongest predictor of belief in climate change and support for climate policies (Todorova et al. 2025).
- Strengthening trust in climate scientists may therefore be an important lever to accelerate climate action.
- A potential loss of trust in climate scientists could have adverse effects on people's engagement in climate action.



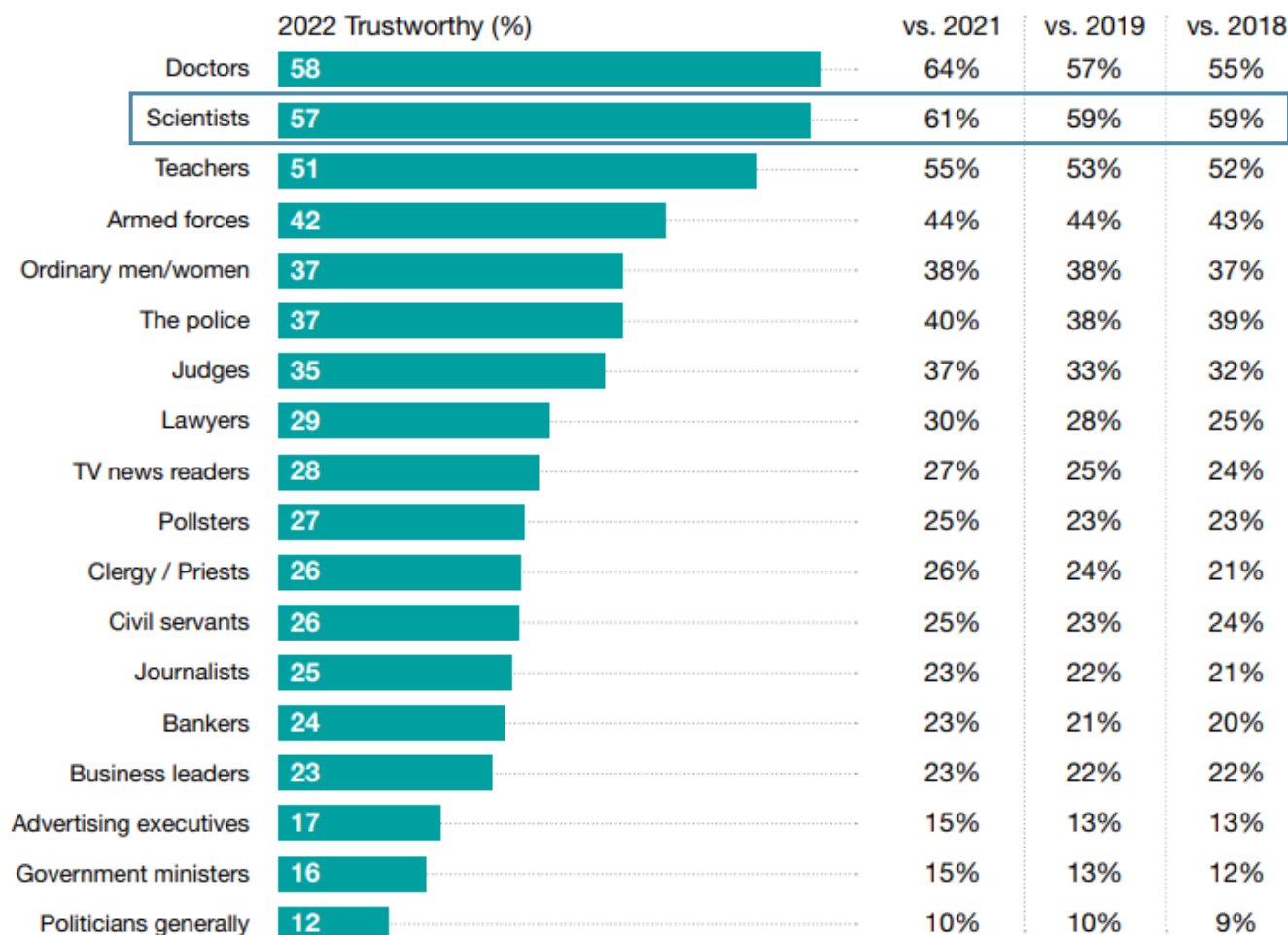
Trust in Climate Science

qualtrics^{XM} + SAP

Globally, trust in science has increased significantly year-over-year, driven in large part by European regions, MENA, Sub-Saharan Africa, LAC and North America. In few regions, the 2021 results exhibit a return to 2019 levels (namely South and East Asia).



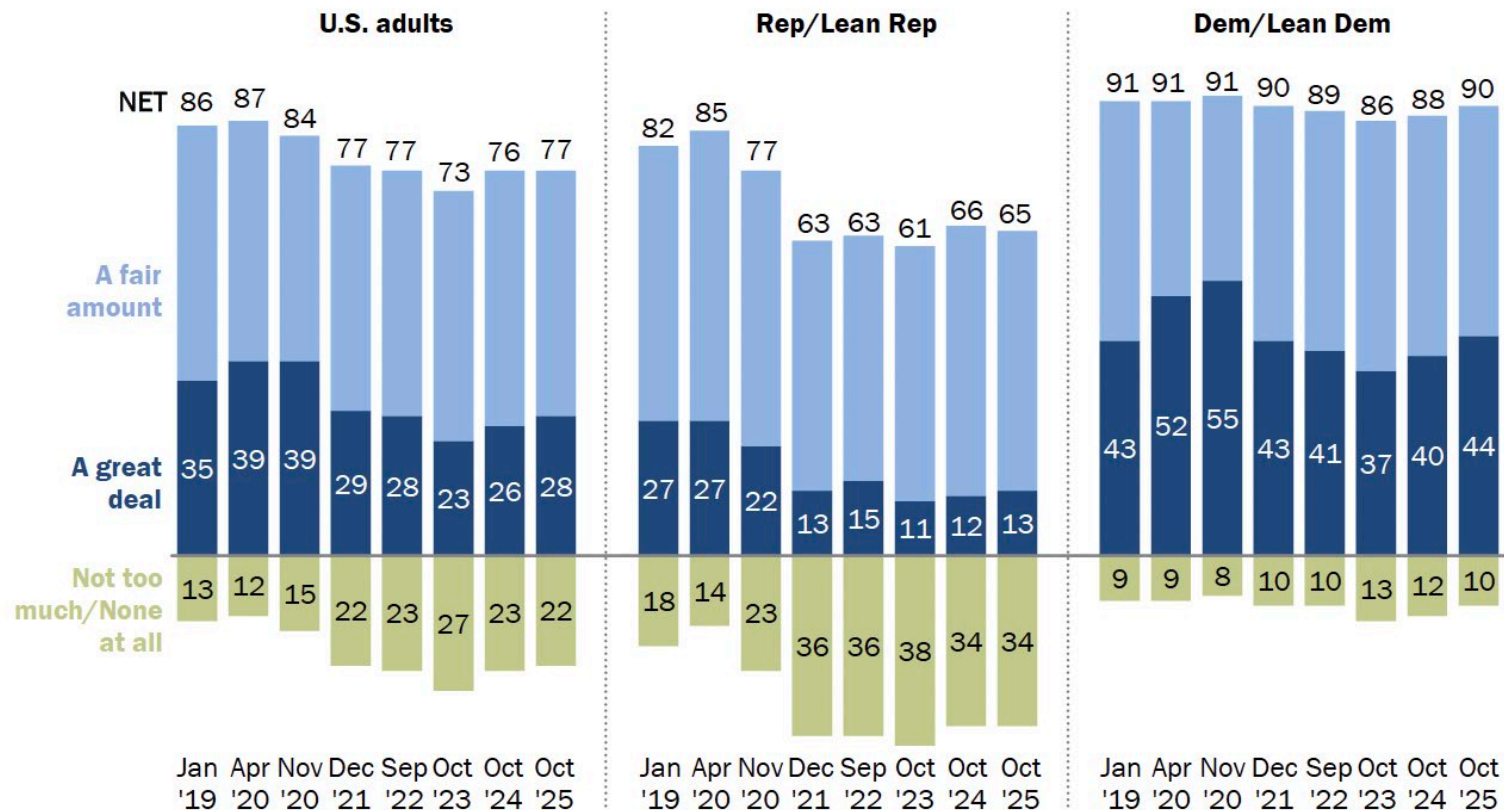
Global trustworthiness (%) trends 2018-2022



Q: Please look at this list of different types of people. In general, do you think each is trustworthy or untrustworthy in your country? **Source:** Ipsos Global Trustworthiness Monitor: Scores presented here are based on a 22-country average covering only those countries which have participated in all previous waves of the survey. The figures differ slightly from the 28-country average presented elsewhere. Note that Russia was included in previous waves but has been omitted from 2022, meaning the 2022 figures do not reflect the same sample as 2021 or previous waves.

Democrats continue to have more confidence in scientists than Republicans

% who have ___ of confidence in **scientists** to act in the best interests of the public

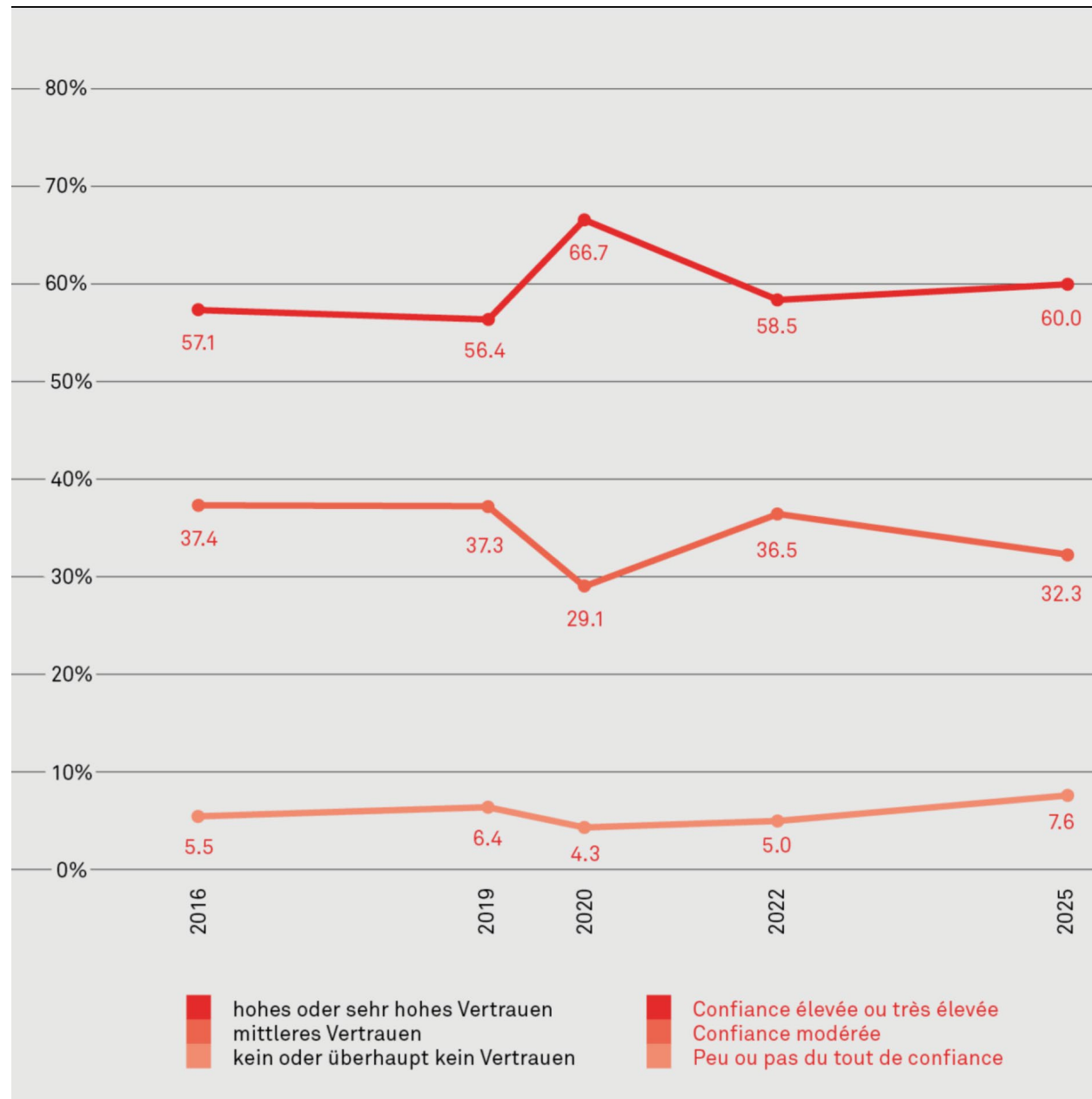


Note: Respondents who did not give an answer are not shown.

Source: Survey of U.S. adults conducted Oct. 20-26, 2025.

"Do Americans Think the Country Is Losing or Gaining Ground in Science?"

PEW RESEARCH CENTER



Trust in Science and Science-Related Populism (TISP)

A Global Many Labs Study
with 241 co-authors at 179 institutions



Viktoria Cologna
Harvard University
Project lead

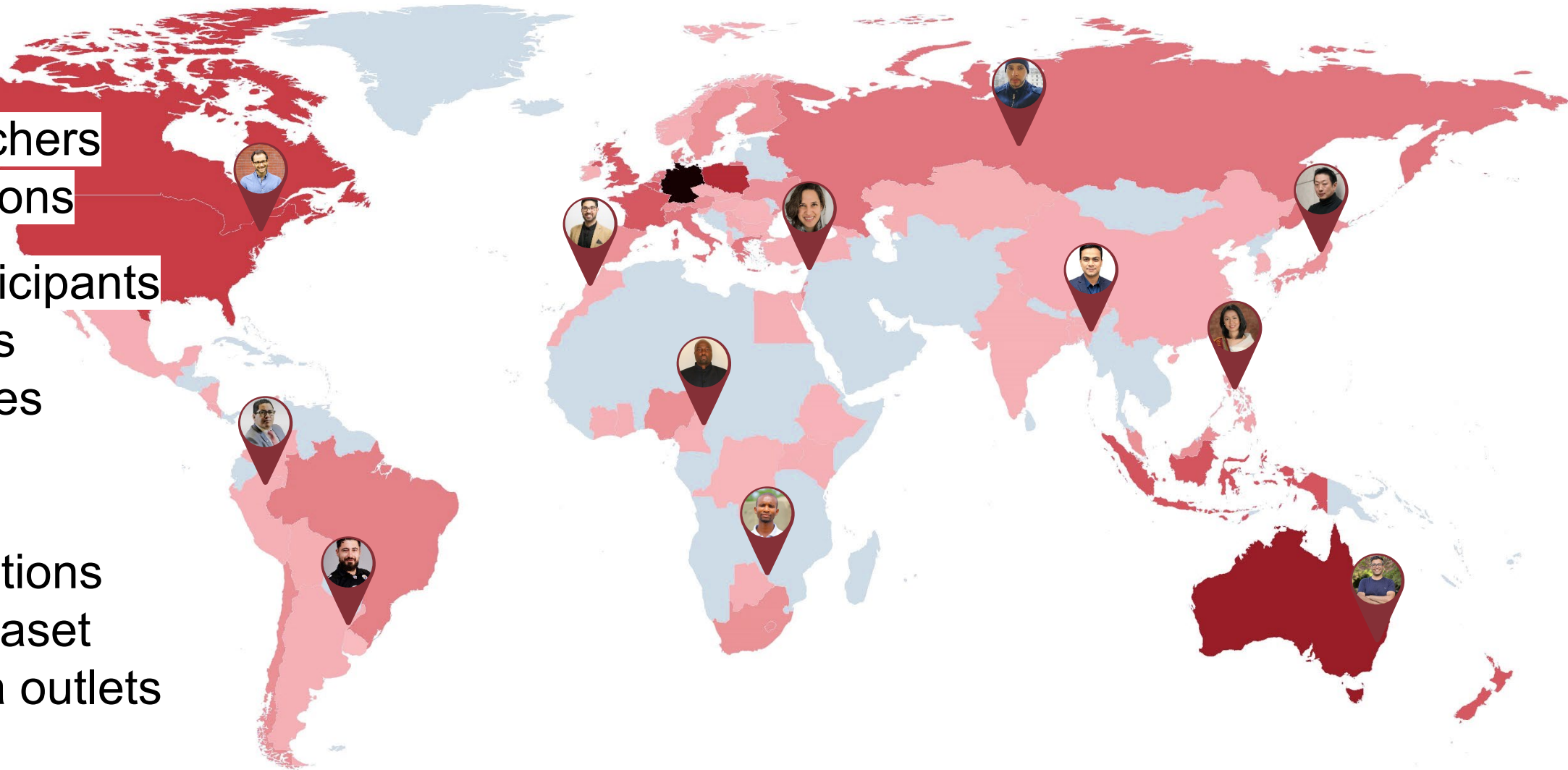


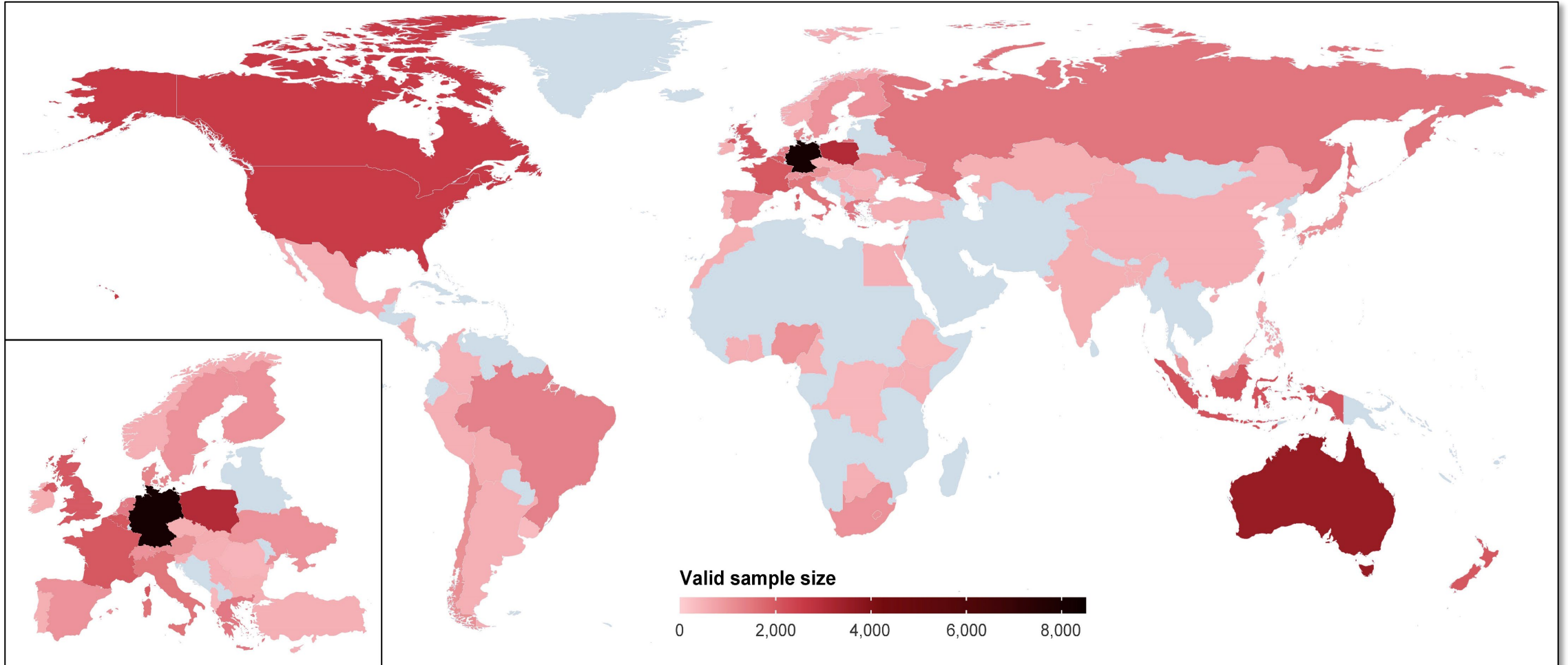
Niels G. Mede
University of Zurich
Second lead

The TISP Project

9 experts
230 researchers
179 institutions
71,922 participants
68 countries
37 languages

3 awards
10+ publications
1 public dataset
400+ media outlets

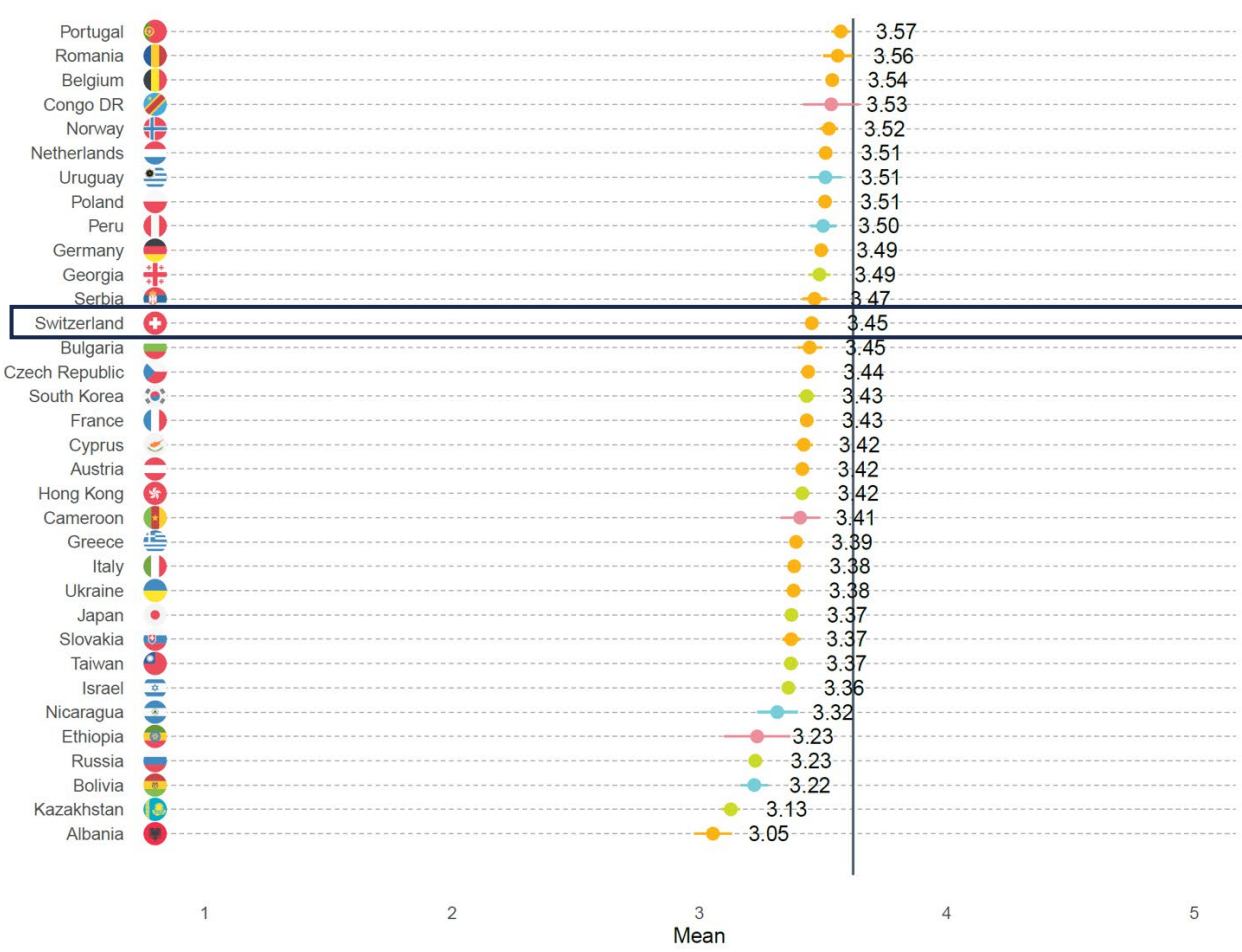
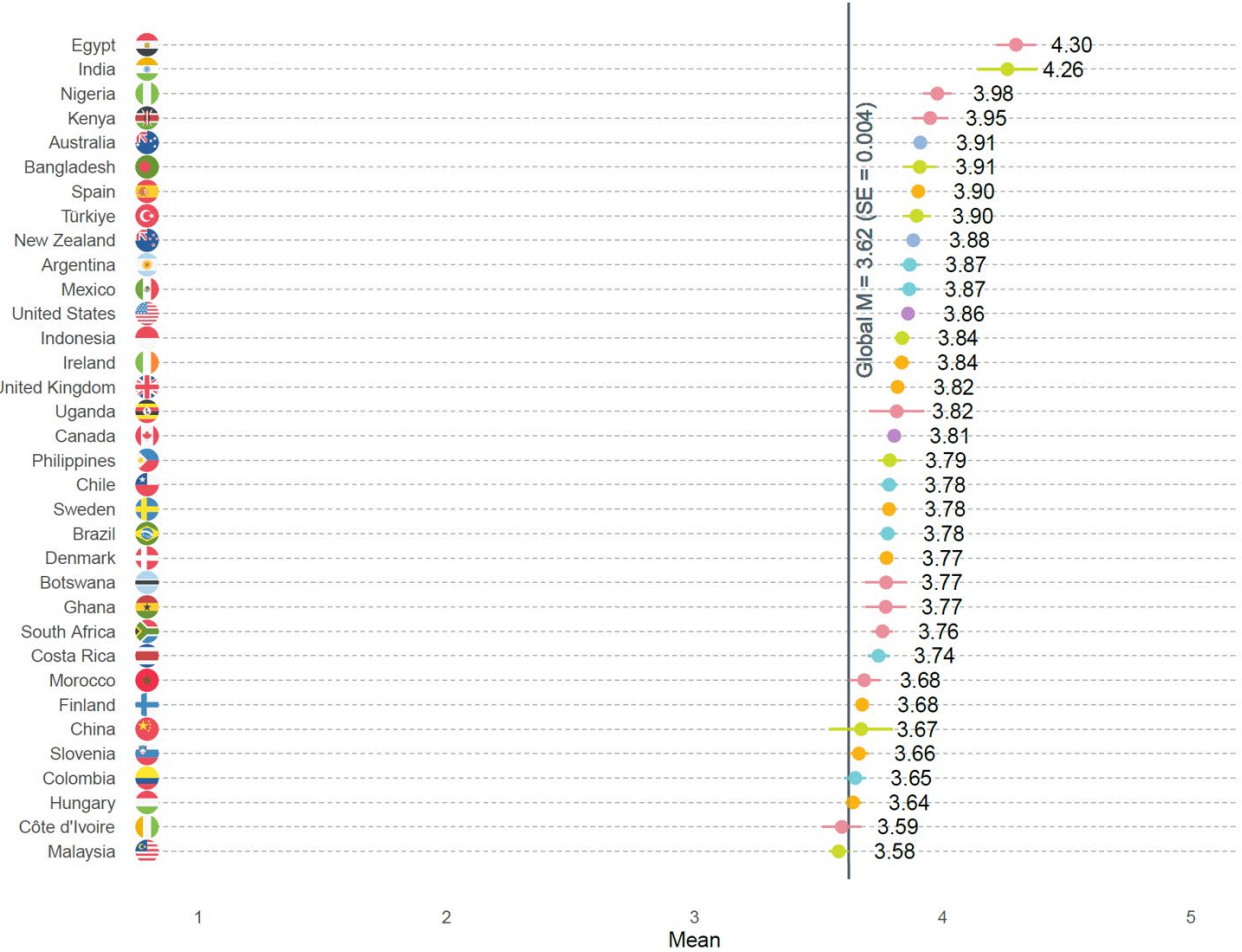




71,922 participants in 68 countries

Trust in scientists around the world

Means and standard errors of trust in scientists across countries
Error bars show standard errors, vertical line indicates global mean



Region Africa Asia Australia/Oceania Europe Latin America North America

Trustworthiness dimensions

1. Competence ($M = 4.02$, $SD = 0.71$)

How expert or inexperienced are most scientists?



2. Integrity ($M = 3.59$, $SD = 0.78$)

How honest or dishonest are most scientists?



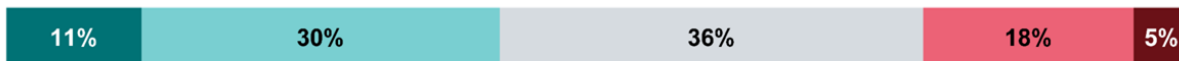
3. Benevolence ($M = 3.55$, $SD = 0.82$)

How concerned or not concerned are most scientists about people's wellbeing?

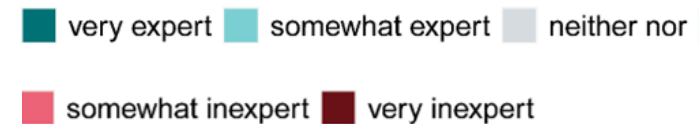


4. Openness ($M = 3.32$, $SD = 0.86$)

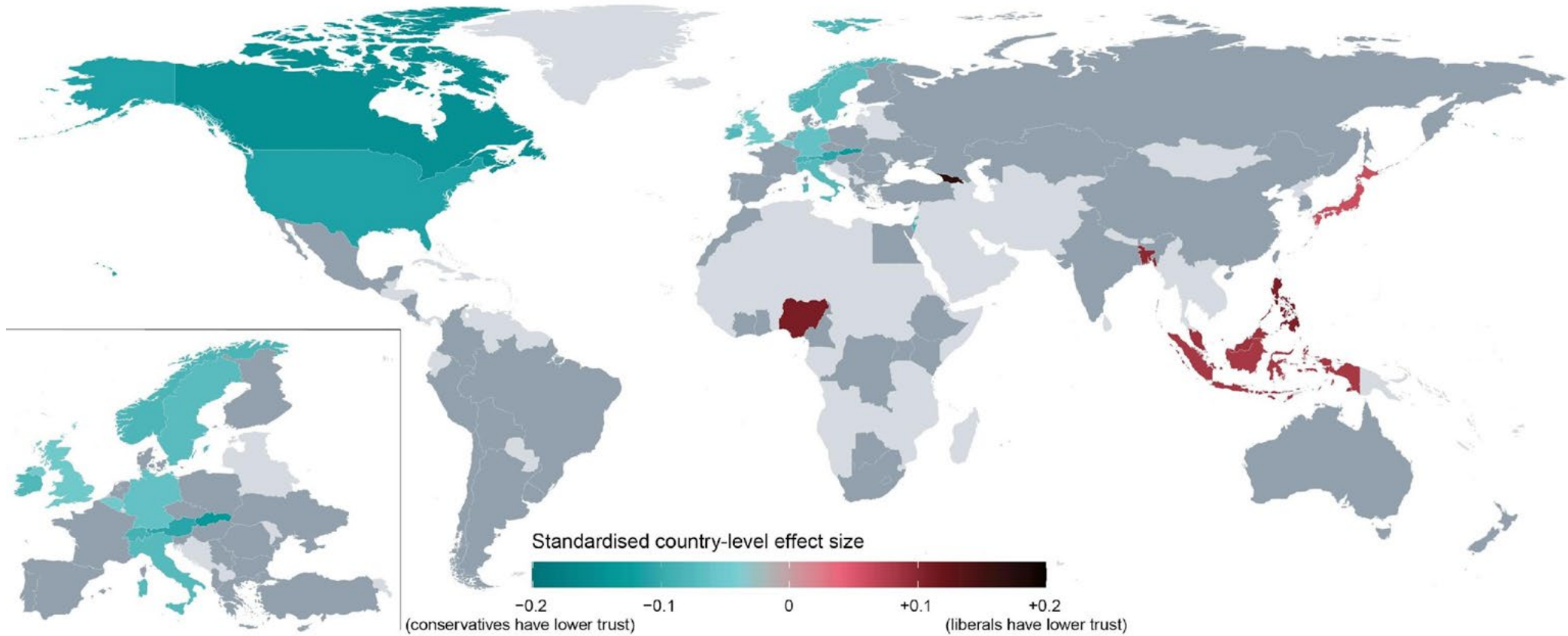
How much or little attention do scientists pay to others' views?



Response options:



Political orientation and trust in scientists



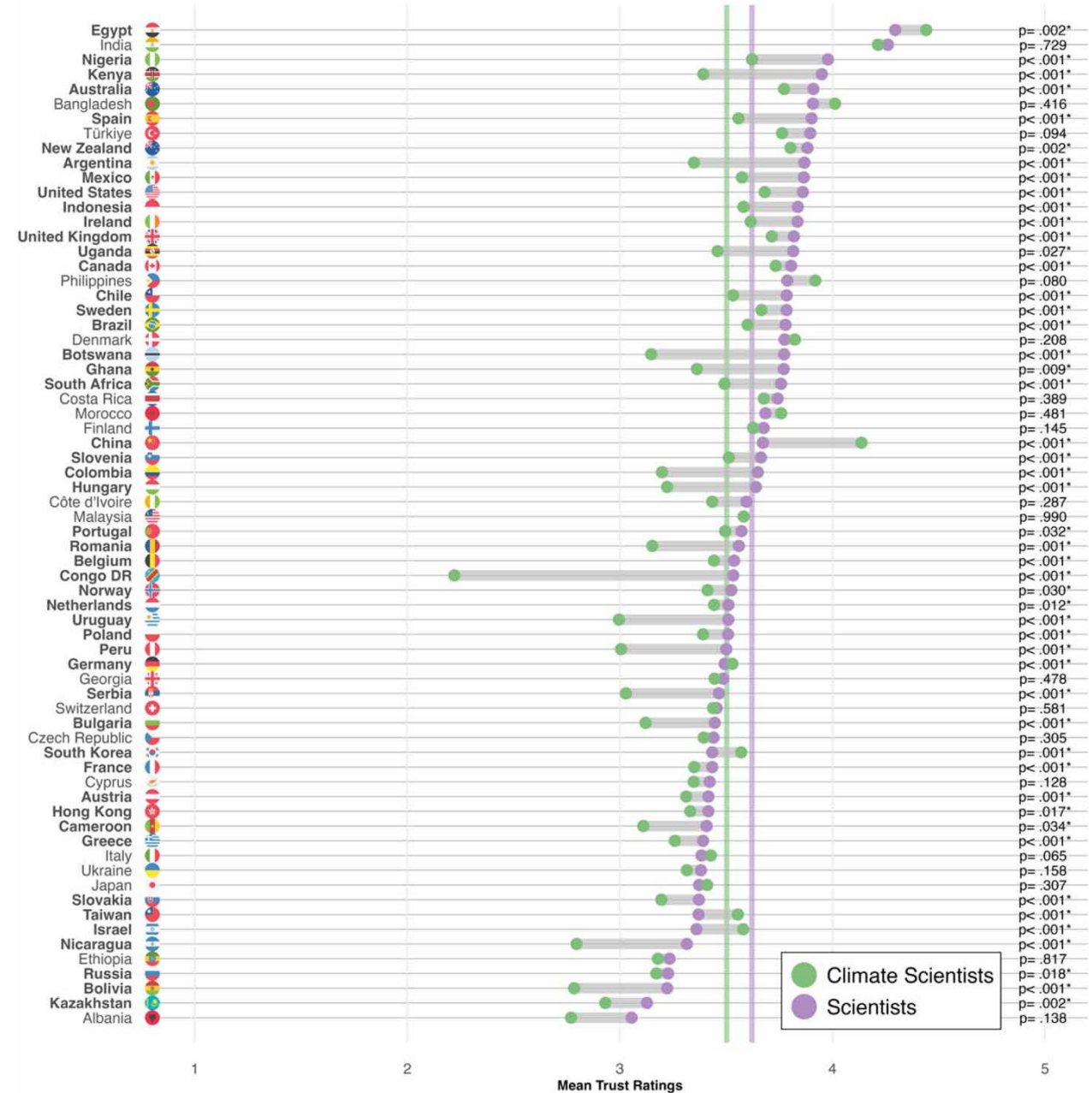
Trust in climate scientists

- Trust in scientists moderately high.

Trust in scientists differs *across disciplines*:

- Climate scientists are less trusted than other scientists (Druckman et al., 2024; Gligorić et al., 2024, 2025)

Ghasemi et al., 2025



Scientists in society and policymaking

Scientists should work closely with politicians to integrate scientific results into policy-making.



Scientists should actively advocate for specific policies.



Scientists should communicate their findings to politicians.



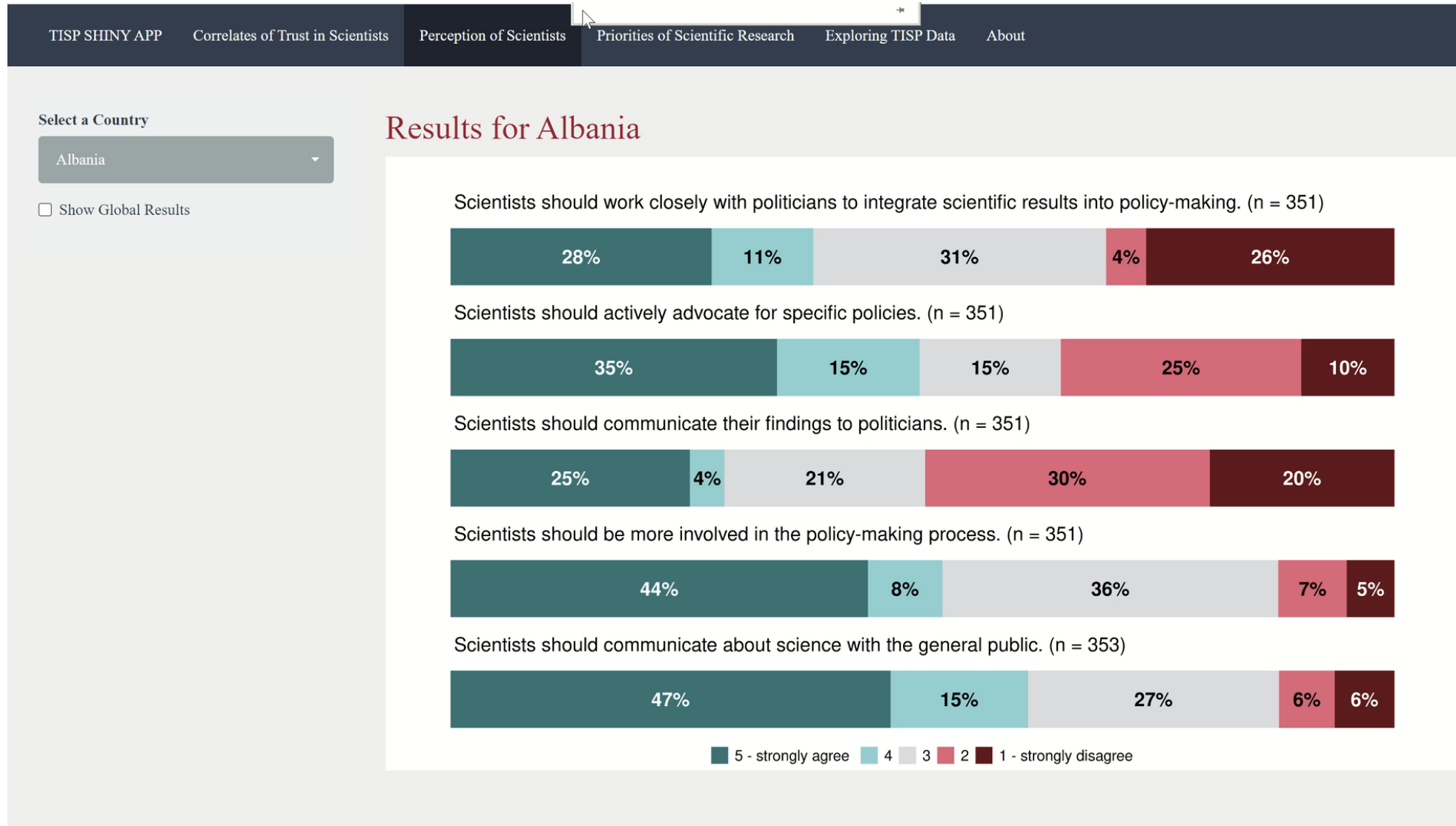
Scientists should be more involved in the policy-making process.



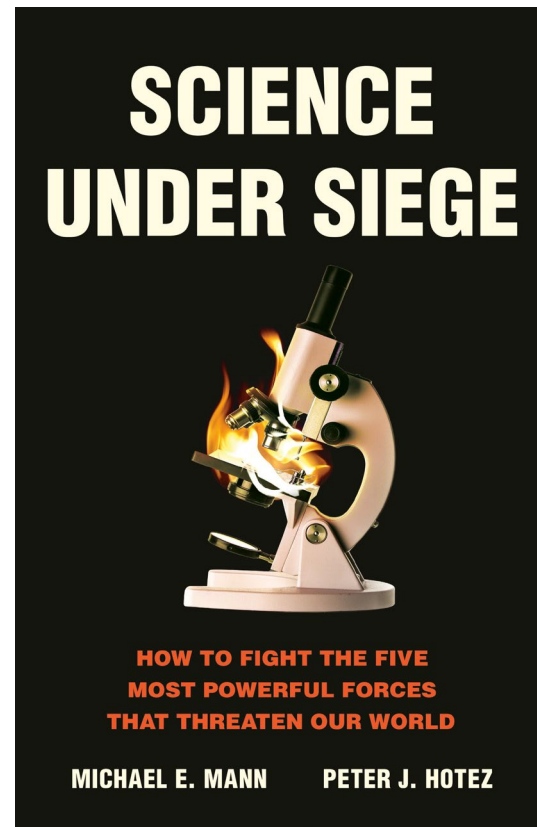
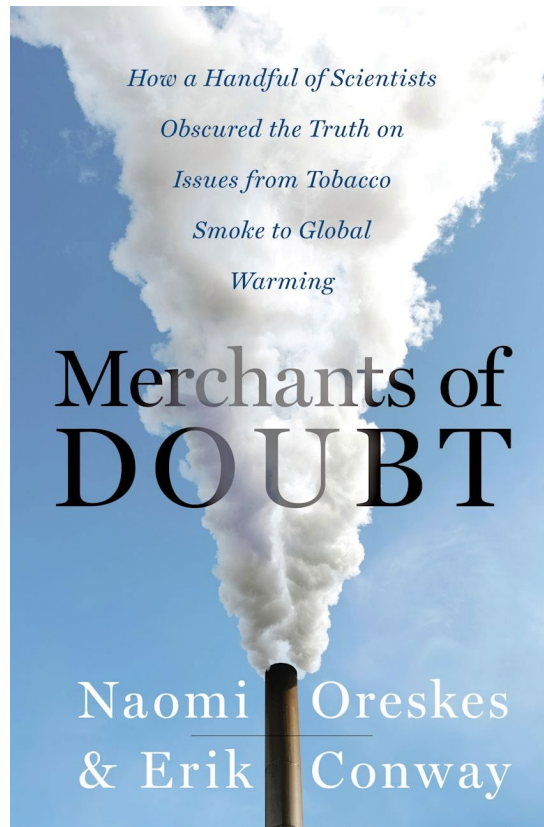
Scientists should communicate about science with the general public.



5 - strongly agree 4 3 2 1 - strongly disagree



“Science under siege”



Trump Administration Plans to Break Up Premier Weather and Climate Research Center

Russell Vought, the White House budget director, called the laboratory a source of “climate alarmism.”

How the Trump administration is already cutting off climate research

President Trump’s efforts to freeze climate spending have sparked warnings of rippling consequences in years ahead. For many climate scientists, the consequences are already here.

July 5, 2025

🔊 8 min ➦ 📌 🗨 269

Which messages are effective at strengthening trust in climate scientists?

Megastudy

“A massive field experiment in which many different treatments are tested synchronously in one large sample using a common, objectively measured outcome” (Milkman et al., 2021)

Intervention tournament:

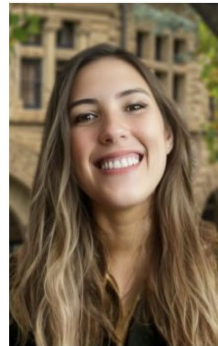
- Posted public call for collaboration
- Selected interventions for testing with an advisory board
- Tested the most promising interventions in a megastudy

Team

Research team



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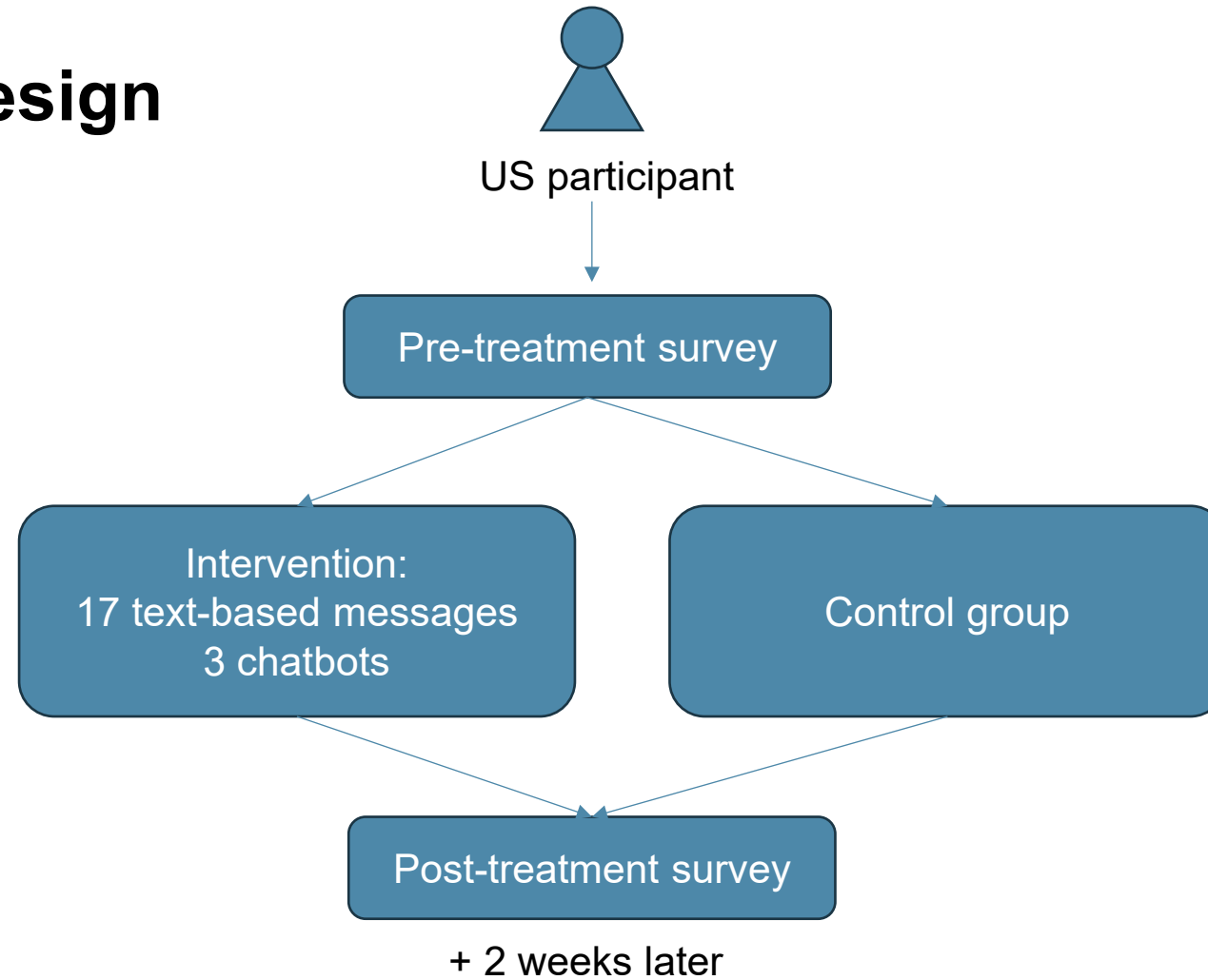


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Jun. Prof. for Science
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(ZPID), *Germany*

Experimental design



Interventions

Collaboration and peer-review

Intervention name	Summary
Interview Prof. Maraun	Climate scientist Prof. Douglas Maraun at the University of Graz in Austria stresses the collaborative and self-correcting process of climate science.
Peer-review	What makes climate science trustworthy is the process of independent peer-review.

Interventions

Scientific methods and results

Intervention name	Summary
Measurement & modeling (1)	Climate scientists use sophisticated measurement and computational modeling techniques to surveil climate and predict how it changes.
Measurement & modeling (2)	Climate scientists are primarily natural scientists (e.g., biologists, physicists). They use sophisticated tools and quantitative methods to measure and predict climate change.
Model accuracy	This is an edited version of a real news article showcasing that even old climate models, despite some flaws, were remarkably correct in predicting global warming.

Interventions

Community & local impact

Intervention name	Summary
Portrait Prof. Cherry	Todd Cherry, a scientist focused on climate issues, is integrated in his local community, and does work that is relevant for this community.
Scientist community helpers	Climate scientists are members of local communities and their work helps local communities in times of climate disasters (e.g., floods and wildfires).

Interventions

Other's endorsement

Intervention name	Summary
Corporate reliance	Insurance companies and large corporations rely on climate scientists' projections.
Former skeptics	Former climate change skeptics Jennifer Rukavina (television meteorologist) and Bob Inglis (former Republican congressman) explain how they came to change their mind.

Interventions

Other types

Intervention name	Summary
Consensus	Correcting potential misperceptions on the level of agreement among climate scientists on climate change.
Oil industry misinformation	Oil companies have spent decades financing large propaganda campaigns to cast doubt on the existence climate change and the credibility of climate scientists.

Example text scientist community helpers

Working together for safer, stronger communities.

Subtitle: Climate science helps people protect what matters most: our families, our nature, and the places we all call home.

Climate scientists are neighbors, parents, and community members. They use tools like weather sensors, soil data, and fire-mapping technology to help our communities prepare for climate change and recover from extreme weather events.

From local farms to coastal towns, scientists and residents work side by side, to keep families safe and livelihoods secure. Their work supports goals shared by everyone: keeping homes safe, land healthy, and future generations protected.

Neighbours helping neighbours

Along the Gulf Coast, many climate scientists live in the towns they study. They volunteer in local shelters, send their kids to the same schools as their neighbors, and worry about the same storms.

During Hurricane Harvey (2017), local scientists worked with emergency managers, meteorologists, and neighborhood volunteers to track rainfall and rising floodwaters.

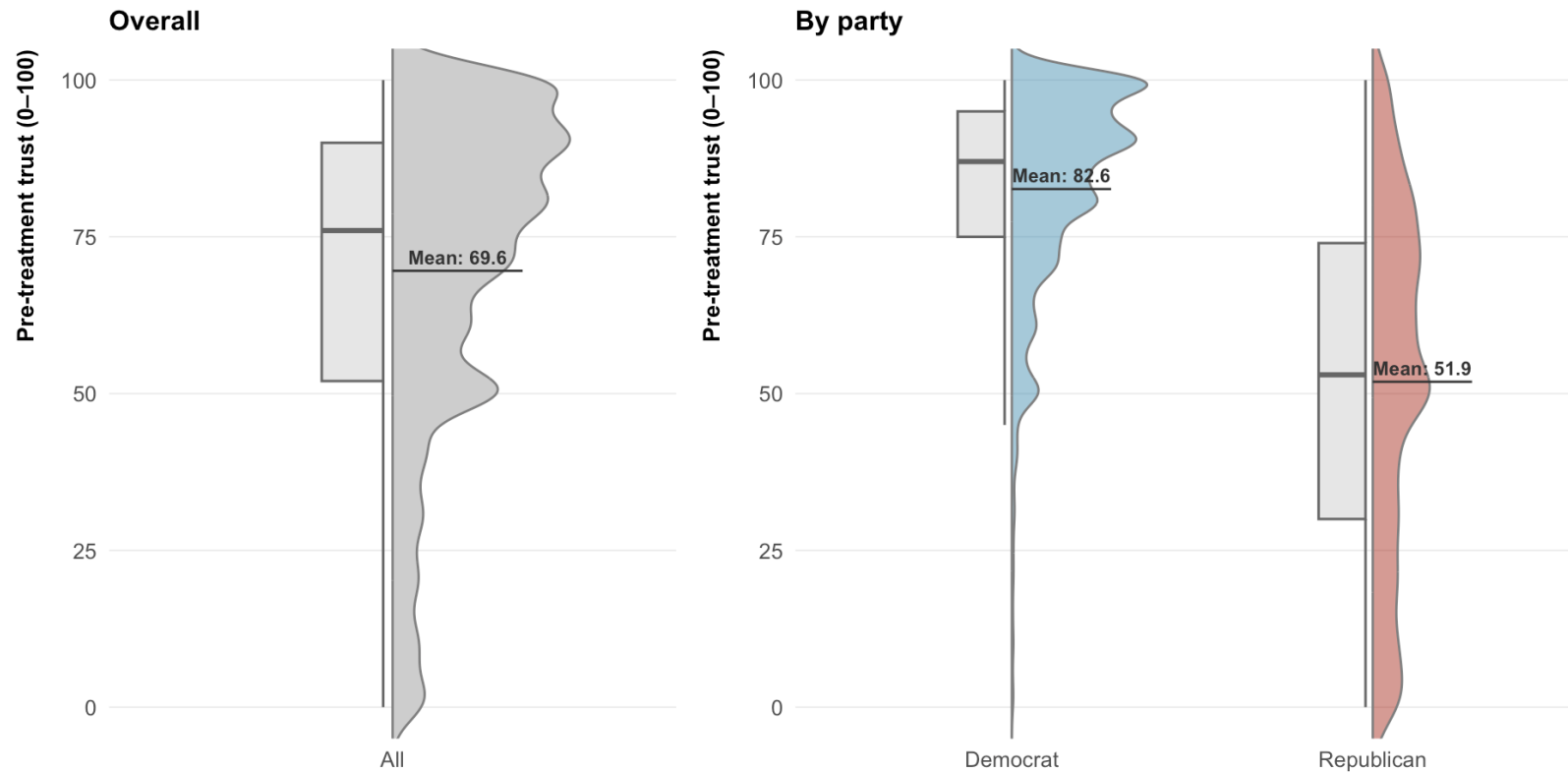
Their real-time flood-risk maps helped firefighters and community groups identify which neighborhoods to evacuate first, and where children, seniors, and hospitals were located.

Climate scientists aren't outsiders looking in – they are community members using their training to help keep families safe.

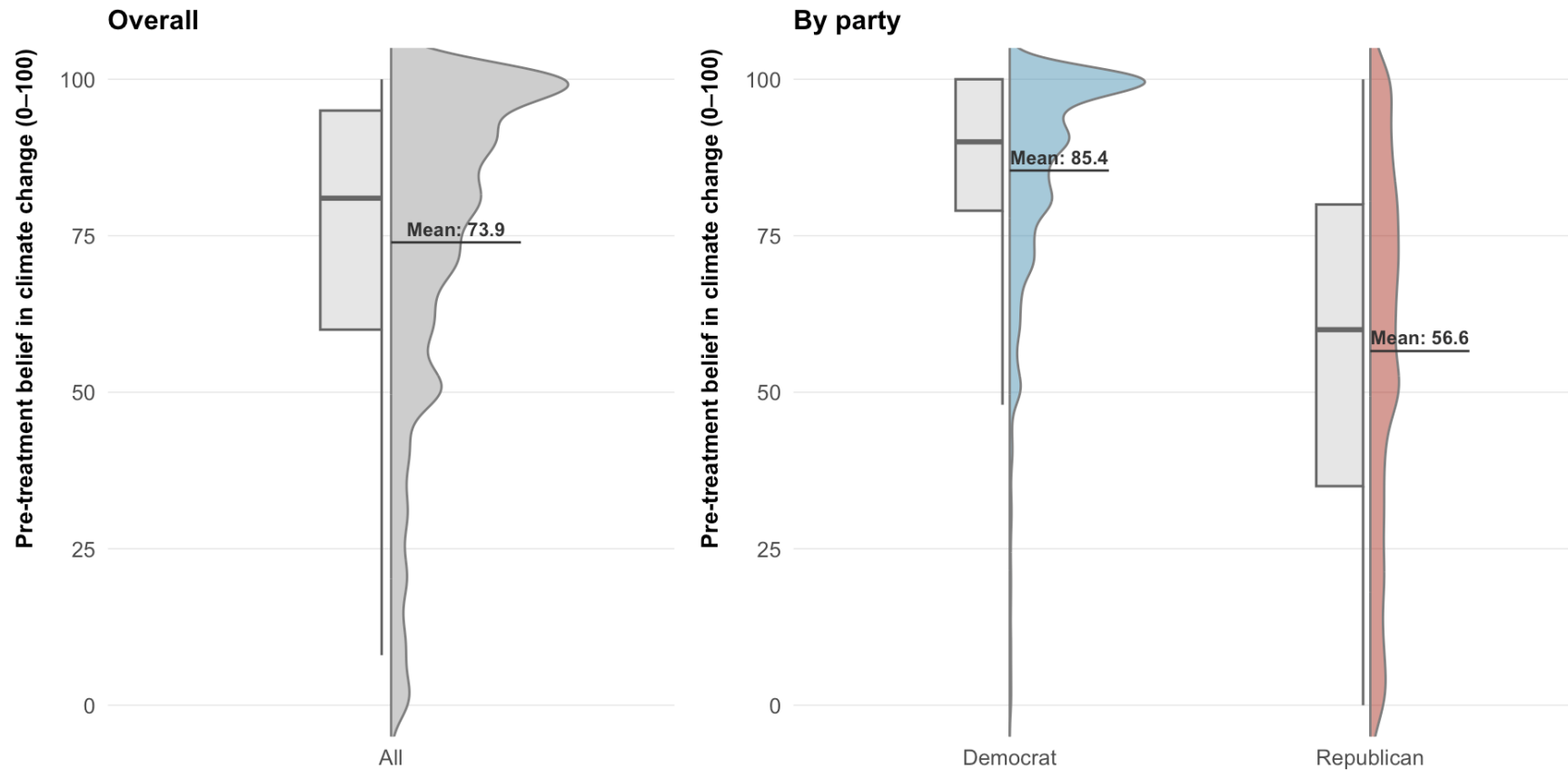
Interventions

Three LLM-chatbots: Participants were invited to have conversations with the chatbots

Trust in climate scientists (N = 26,951)

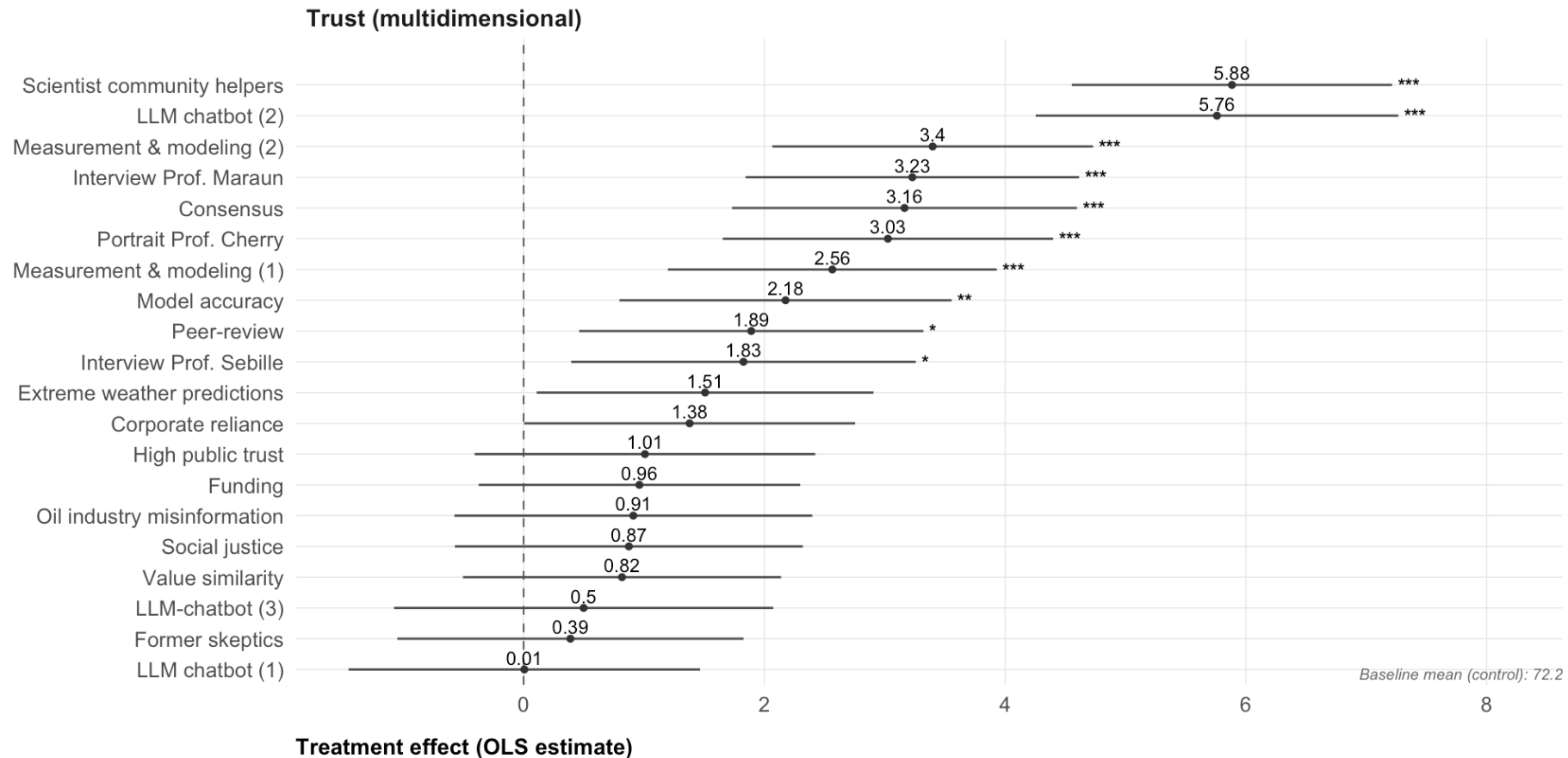


Trust in climate scientists (N = 26,951)

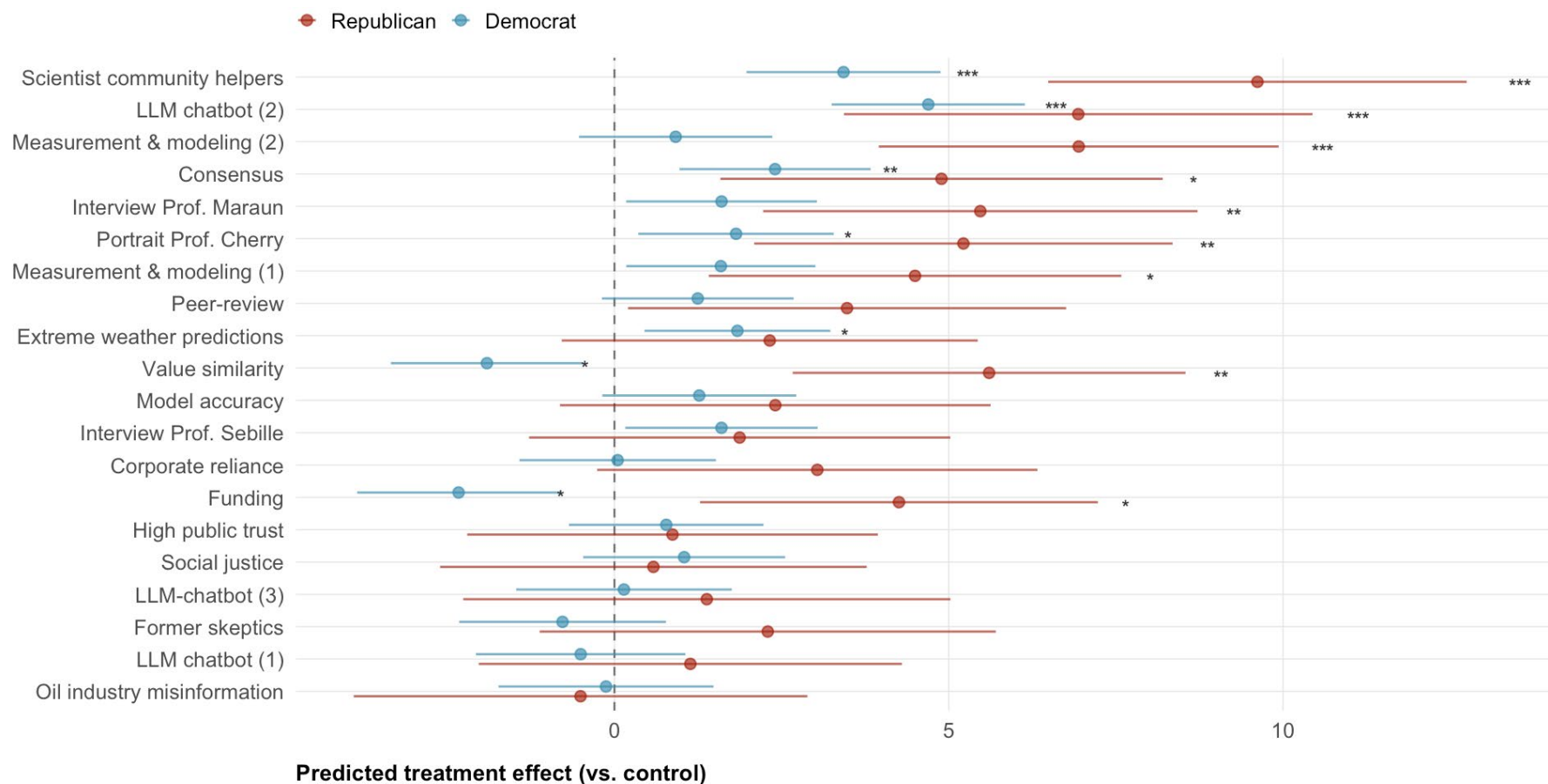


Intervention effectiveness

Effects on trust in climate scientists



Effects on trust by political orientation



No effects on...

- Climate change beliefs
- Behavioral intentions

→ Trust is necessary but not sufficient

Summary of findings

- 10 out of 20 interventions significantly increased trust in climate scientists
 - Especially those focusing on:
 - how sophisticated methods in climate science are
 - local engagement of climate scientists
 - personal stories of (real) climate scientists
- Effects for chatbots are mixed
- Mostly small effects, but some sizeable among Republicans

Recommendations for climate communicators

1

Find common ground and talk with, not at, your audience

Try to find and occupy shared ground, instead of speaking from the high ground. Find a way to show that you are on the side of the people you're talking to.

Benevolence

This isn't just about meeting people where they're at or even endorsing or repeating unhelpful ideas or inaccurate beliefs just to create an immediate connection.

It's about finding something or several things that allow you to connect with an audience on a deeper level. It could be a personal story that people will relate to. Or a widely shared belief or motivation, like the importance of passing on a healthy planet to children and grandchildren.

CLIMATE OUTREACH • Beyond 'trusted messengers'



Image credit: SolStock

30

Convey your passion and motivation as well as your credibility

2

oo



Image credit: 10'000 Hours

Influential people are described as passionate and inspiring.

Expertise & Integrity

Audiences will trust you if they think you're genuine, and you know at least a little bit about what you're talking about.

For most people pure passion doesn't trump lack of knowledge. But equally expertise doesn't influence by itself.

CLIMATE OUTREACH • Beyond 'trusted messengers'

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3

Make sure the messenger fits the message (and vice versa)

Are you the right person to address this audience?

The context and particular circumstances affect whether or not people trust a messenger in any given moment. For instance, a famous tennis player is unlikely to be a trusted spokesperson on public transport schemes if they can't remember the last time they were on a bus or a train.

When designing communications strategies, try to draw on existing positive associations that people will have between a particular issue and a relevant communicator. For instance, a local restaurateur or chef can talk about the appeal and scope of a planet-friendly diet more authentically than a business liaison officer with no connection to food.

CLIMATE OUTREACH • Beyond 'trusted messengers'



Image credit: SolStock

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Conclusion

- Communication strategies are likely to fail if they don't come from a trusted messenger.
- The messenger effect underscores the importance of strategically choosing messengers who resonate with different audience segments (e.g., weathercasters)
- Climate scientists are among the most trusted actors, and sharing personal stories of climate scientists can help to increase trustworthiness.